Measurement of Scintillating Fibres for the LHCb Experiment

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Abstract

This report investigates the physical properties and light transport mechanisms of scintillating fibres used in the LHCb SciFi Tracker. Through a combination of Monte Carlo simulations (Geant4) and experimental measurements using a spectrometer and an LED source, we characterize the angular distribution of emitted light and its attenuation length. By comparing experimental results with simulated data, we aim to validate our theoretical understanding of the fibres' behavior.

1 Introduction

Tracking detectors are fundamental in particle physics for reconstructing the trajectories of charged particles. Scintillating fibres (SciFi) represent a versatile technology offering high spatial resolution and efficient photon detection. In the context of high-energy physics (HEP), these technologies are prominently used as active materials in modern detectors, most notably within the LHCb experiment at CERN. The specific role of the SciFi Tracker in the detector system, as well as its underlying physical principles, will be described in greater detail in the following sections.

1.1 The LHCb Experiment

The LHCb detector is one of the four primary experiments at the Large Hadron Collider (LHC) at CERN, in Geneva, where protons are accelerated and collided at centre-of-mass energies up to $\sqrt{s} = 13.6\,\text{TeV}$. Designed as a single-arm forward spectrometer, it performs precision measurements in the b- and c-hadron sectors [1]. The spectrometer system, as shown in Figure 1, includes the Vertex Locator (VELO), the Upstream Tracker (UT), the dipole magnet, and the Scintillating Fibre (SciFi) Tracker. Together, these components precisely reconstruct charged particle trajectories and measure their momenta. Particle identification, essential for distinguishing long-lived hadrons (e.g., pions, kaons, protons), is provided by two Ring Imaging Cherenkov detectors (RICH1 and

RICH2), which operate at different ranges of particles' momenta. These detectors determine particle velocities by measuring the emission angle of Cherenkov radiation[1]. The emission angle of the resulting light cone directly correlates to the particle's velocity. Combining this velocity with the independently measured momentum allows for precise mass calculation and particle identification. Furthermore, the calorimeter system, comprising electromagnetic (ECAL) and hadronic (HCAL) components, measures particle energies by detecting the total energy deposited in the medium, via the development of electromagnetic and hadronic showers. Finally, the muon stations (M2–M5) are located at the end of the detector. Since muons do not feel the strong interaction and experience suppressed bremsstrahlung compared to electrons (due to their higher mass), they are able to traverse the entire detector without initiating showers, making these outer chambers crucial for their successful detection.

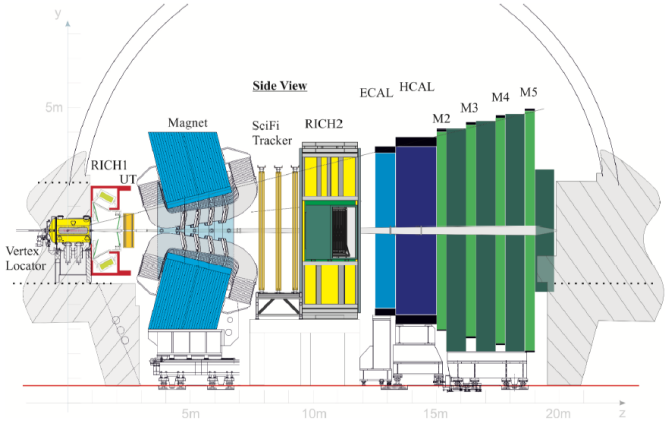


Figure 1: Schematic overview of the LHCb detector and its subsystems after the upgrade for Run 3.

1.2 The SciFi Tracker

During the major upgrade of the LHCb detector (Upgrade I), the Scintillating Fibre (SciFi) Tracker (Figure 2) was introduced to replace the previous Inner and Outer Trackers, whose granularity was insufficient for the increased track multiplicities [2]. Located downstream of the magnet, the SciFi Tracker is indispensable for reconstructing the trajectories of charged particles bent by the magnetic field, achieving a spatial resolution of less than 100 μm. The tracking system consists of three stations (T1, T2, and T3). Each station comprises four layers arranged in an $x - u - v - x$ geometry, where the u- and v-layers are

[1] Cherenkov radiation is emitted when a charged particle passes through a medium with a certain rifraction index at a speed greater than the velocity of light in that medium.

rotated by $\pm 5^\circ$ around the beam axis. This angle configuration allows the detector to provide additional information on the vertical position of the particles. The active material of the detector consists of double-clad scintillating fibres with a total diameter of 250 μm [3]. Each fibre is composed of a polystyrene core (diameter 220 μm) and two surrounding cladding layers (each 7.5 μm thick). The refractive indices decrease outwardly from the core ($n = 1.60$) to the inner cladding ($n = 1.49$) and the outer cladding ($n = 1.42$), enabling the trapping and propagation of scintillation light via total internal reflection. Macroscopically, the fibres are wound into dense mats composed of six staggered layers. Eight of these mats are then combined to form larger modules. To maximize light collection, one end of each fibre mat is equipped with a reflective mirror, while the opposite end is read out by arrays of Silicon Photomultipliers (SiPMs), which detect the arriving photons and convert them into electrical signals.

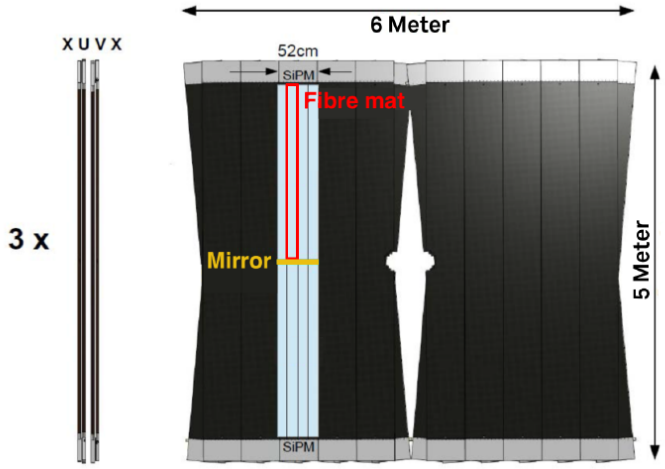


Figure 2: Schematic view of the SciFi Tracker. Left) Side of the image is a side view of a station consisting of four layers. Right) Front view of the SciFi Tracker, with the two halves slightly separated with the beam pipe hole at the center. A single fibre mat is highlighted in red, while the reflective mirror is shown in yellow.

2 Physical Principles

2.1 Scintillation Mechanism and Spatial Resolution

The scintillation mechanism employed in the LHCb SciFi Tracker is basically the same as all scintillating fibers. When an ionising particle traverses the fiber, the valence electrons within the polystyrene core are excited. The subsequent non-radiative relaxation to the lowest vibrational state is followed by the emis-

sion of a UV photon. To enhance the relatively poor intrinsic quantum yield of pure polystyrene, a primary dye (p-terphenyl) is added. However, photons emitted by p-terphenyl suffer from a short attenuation length of approximately 1 m. Thus, a secondary wavelength-shifting dye, TPB (tetraphenyl-butadiene), is incorporated to absorb and re-emit these photons at longer wavelengths, effectively increasing the attenuation length to the required > 3 m.

Despite the fibres having a relatively large total diameter of $d = 250\,\mu$m, the tracking system successfully achieves a spatial resolution of less than $100\,\mu$m. This can be demonstrated mathematically by modeling the particle hit position as a continuous uniform distribution across the active width of the fibre. The spatial resolution is defined by the standard deviation σ of this distribution, which is the square root of its variance:

$$\sigma = \sqrt{\frac{d^2}{12}} = \frac{250\,\mu\text{m}}{\sqrt{12}} \approx 72.2\,\mu\text{m} \tag{1}$$

This theoretical resolution of $72.2\,\mu$m explains how the sub-$100\,\mu$m precision is achieved experimentally.

2.2 Photon Transport and Total Internal Reflection

The generated photons are guided along the fibre via total internal reflection at the interfaces between the core and the claddings. The refractive indices decrease outwardly to facilitate this: $n_{core} = 1.60$, $n_{clad1} = 1.49$, and $n_{clad2} = 1.42$. For meridional photons (those crossing the central longitudinal axis of the fibre), the maximum emission angle θ_{max} relative to the fibre axis that still undergoes total internal reflection can be derived using Snell's Law. At the boundary, the critical angle with respect to the normal is defined as $\theta_c = \arcsin(n_{clad}/n_{core})$. Since the emission angle θ relative to the fibre axis is the complement of the angle to the normal ($\theta = 90^\circ - \theta_c$), we obtain:

$$\cos\theta_{max} = \frac{n_{clad}}{n_{core}} \implies \theta_{max} = \arccos\left(\frac{n_{clad}}{n_{core}}\right) \tag{2}$$

Applying this geometric relation to the two optical interfaces yields the theoretical maximum angles:

$$\theta_{max1} = \arccos\left(\frac{1.49}{1.60}\right) \approx 21.4^\circ \tag{3}$$

for the core to inner-cladding interface, and:

$$\theta_{max2} = \arccos\left(\frac{1.42}{1.60}\right) \approx 27.4^\circ \tag{4}$$

for the inner to outer-cladding interface. These theoretical limits mathematically govern the geometric acceptance of the fibres and represent key thresholds that are later investigated in the experimental angular distributions.

3 Setup and Methods

3.1 Experimental Setup

The laboratory setup (Figure 3) consists of a single scintillating fibre, a mobile LED light source, and a spectrometer. In the actual LHCb SciFi Tracker, readout is performed by Silicon Photomultipliers (SiPMs) operated at $-40°$C to suppress thermally induced noise. However, because SiPMs measure the integrated light yield and cannot provide information regarding the wavelength or the emission angle of the incident photons, our setup employs a spectrometer to investigate these specific dependencies. The LED is mounted on a motorized device that moves along the fibre's longitudinal axis (x-axis). The spectrometer is equipped with two additional motors, enabling it to scan a specific range of horizontal (-20°,90°) and vertical (-6°,90°) angles around the fibre's end. All motors, along with the LED and the data acquisition system, are interfaced with a local computer and controlled via a dedicated program developed specifically for this laboratory experience.

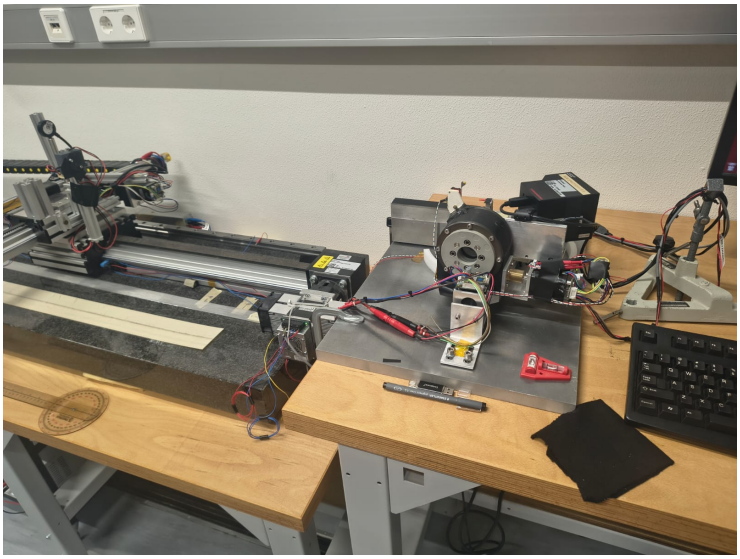


Figure 3: Photo of the experimental setup used during the experience: on the left the LED source with its support located along the longitudinal motor; on the right side the spectrometer and its two angular motors. Everything is wired and connected to the local computer.

3.2 Simulation Overview

Before detailing the analysis procedures, it is necessary to introduce the raw simulation dataset generated via Geant4. The software models the propagation

of individual photons through 50 distinct fibres. Each fibre is excited at 24 longitudinal positions, ranging from 100 mm to 2400 mm from the readout end, with a spacing of 100 mm. A simulated detector volume is placed at the fibre's end to register the arriving photons and store their physical parameters.
For each detected photon, the dataset records the following key variables:

- **Position Coordinates:** `x_start`, `y_start`, and `z_start` denote the creation coordinates, while `y_exit` and `z_exit` record where the photon exits the fibre.
- **Initial Momentum:** `px_start`, `py_start`, and `pz_start` represent the normalized components of the momentum direction vector at creation ($p_x^2 + p_y^2 + p_z^2 = 1$).
- **Interactions:** `reflCoCl` and `reflClCl` indicate the number of reflections at the core-cladding and cladding-cladding interfaces, respectively . The variable `rayleighScatterings` tracks the number of Rayleigh scattering events experienced by the photon.
- **Path and Source:** `length_core` and `length_clad` measure the distance traveled within each respective material, `wl` stores the photon's wavelength, and `gpsPosX` represents the x-coordinate of the primary particle responsible for the fibre's excitation.

4 Data Analysis and Results

4.1 Spectrometer Measurement

Firstly, the spectrum of intensity was measured by a spectrometer by setting different variables. For this section, we performed measurements of the light received from the spectrometer with the room light turned on and off, and then we subtracted the background (dark counts) from the actual intensity (counts). This allows for the filtering of environmental photons, resulting in a "cleaned" intensity spectrum.

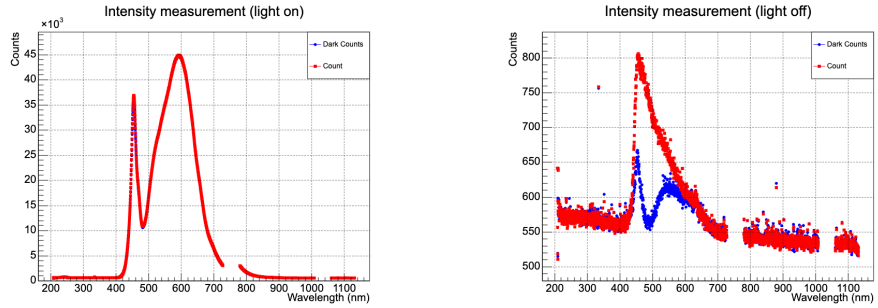


Figure 4: Intensity measured from the spectrometer against the wavelength with the room light on (Left) and off (Right). The total photon count is highlighted in red while the dark counts (photons measured when the LED source was off) are shown in blue.

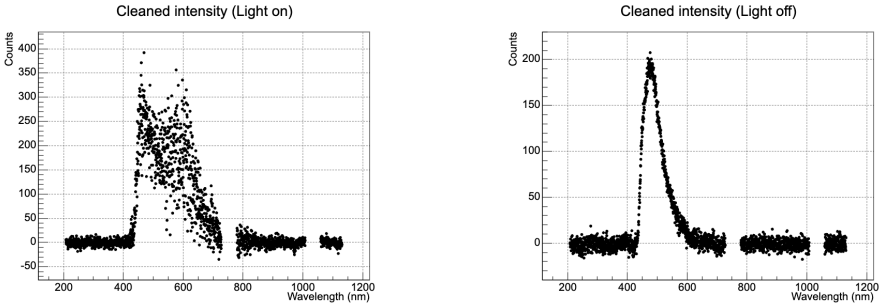


Figure 5: Intensity measured from the spectrometer against the wavelength with the room light on (Left) and off (Right), after subtracting the dark counts. The total photon count is highlighted in black.

As observed, when the room lights were switched on, the intensity measurement was disrupted because the ambient light was detected by the sensor as a significant noise signal. In contrast, when the room lights were switched off, the sensor primarily detected the light from the source, resulting in a much clearer histogram. However, minor fluctuations are still present in the spectrum due to residual ambient light from a 3D printer that was running in the room during the data acquisition.

4.2 Radial Symmetry

The measurement was repeated by recording the intensity at various horizontal ($-18°$ to $30°$, in $3°$ increments) and vertical angles ($-6°$ to $30°$, in $3°$ increments)

to investigate the radial symmetry of the light emission. The resulting two-dimensional distribution is presented in Figure 6.

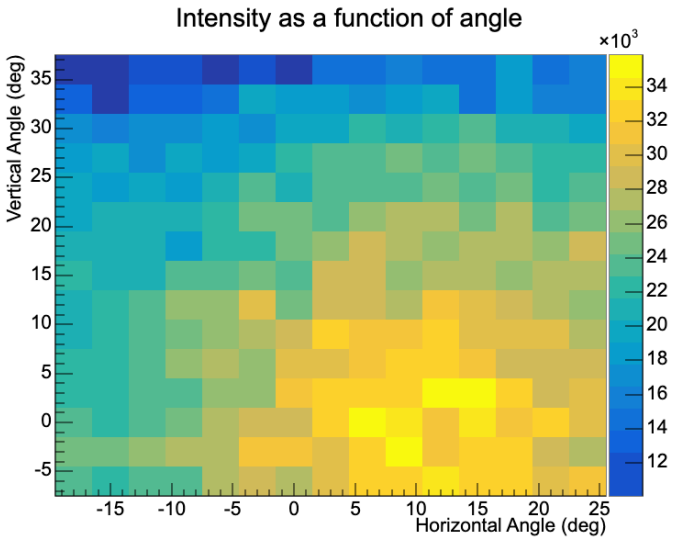


Figure 6: Two-dimensional histogram of the integrated intensity as a function of the horizontal and vertical angles.

As observed in the heatmap, the maximum intensity is not centered at $(0°, 0°)$ as theoretically expected. This asymmetry towards positive horizontal angles (the center is around $\sim +12°$) is likely due to a misalignment in the experimental setup, where the spectrometer's reference frame was not perfectly centered with respect to the optical axis of the fibre.

4.3 Simulation Analysis

To gain deeper insights into the photon transport mechanisms, we analyzed a dataset generated via Monte Carlo simulations in Geant4. The simulation models 50 individual fibres, each excited at 24 different longitudinal positions ranging from 100 mm to 2400 mm from the readout end. To ensure the physical validity of the dataset, unphysical events and specific scatterings were filtered out prior to the analysis. Specifically, all photons that underwent Rayleigh scattering were removed. Furthermore, the distance from the fibre center at the exit point was calculated as $r_{exit} = \sqrt{y_{exit}^2 + z_{exit}^2}$. Any photon with an exit distance exceeding the geometric bound of the fibre ($r_{exit} > 0.125\,\text{mm}$) was discarded (Figure 7).

The remaining photons were categorized into core and cladding photons based on their reflection history. Photons reflecting strictly at the core-inner cladding interface were labeled as core photons, whereas those reaching and reflecting at the interface between the two cladding layers were classified as

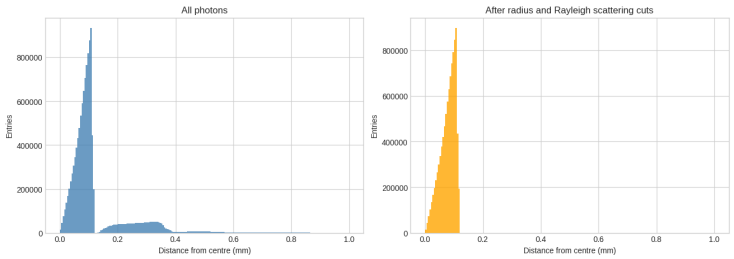


Figure 7: Radial distance of photons from the fibre center before (left) and after (right) data cleaning. Unphysical simulation artifacts beyond the physical fibre radius (0.125 mm) and photons subjected to Rayleigh scattering have been successfully filtered out in the right panel.

cladding photons. A crucial parameter for this analysis is the emission angle θ relative to the fibre's central axis. Since the initial momentum vector is normalized ($p_x^2 + p_y^2 + p_z^2 = 1$), the longitudinal angle can be directly extracted as:

$$\theta = \arccos(p_x) \tag{5}$$

To investigate how the initial trajectory determines whether a photon is confined within the core or penetrates into the cladding, the angular distributions of θ for both categories were histogrammed and overlaid, as shown in Figure 8.

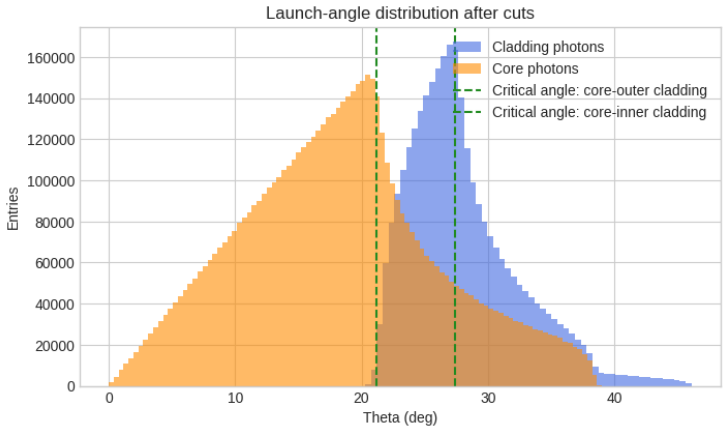


Figure 8: Angular distribution of the simulated photons, separated into core (orange) and cladding (blue) photons. The vertical dashed green lines indicate the theoretical maximum trapping angles for meridional rays at the core-inner cladding and core-outer cladding interfaces, respectively. The high-angle tails extending beyond these theoretical limits represent the contribution of skew rays ($r_{min} > 0$) propagating through the fibre.

The simulated angular distribution confirms theoretical expectations. For both core and cladding photons, the counts exhibit a linear increase from 0° up to their respective critical angles ($\theta_{max1} \approx 21.4°$ and $\theta_{max2} \approx 27.4°$). This linear rise is dictated by the increasing geometric acceptance of the fibre. The distinct presence of photons extending beyond these theoretical maximum limits (marked by the dashed lines) is a direct consequence of non-meridional photons. Since these photons do not cross the fibre's central axis ($r_{min} > 0$), they follow helical trajectories and strike the interfaces at more grazing incident angles, allowing them to be trapped even at steeper longitudinal angles. Furthermore, analyzing the angle θ as a function of the minimum distance from the center (Figure 9) undernlines that photons propagating along helical trajectories (higher r_{min}) can remain trapped at angles significantly higher than those permitted for purely meridional trajectories.

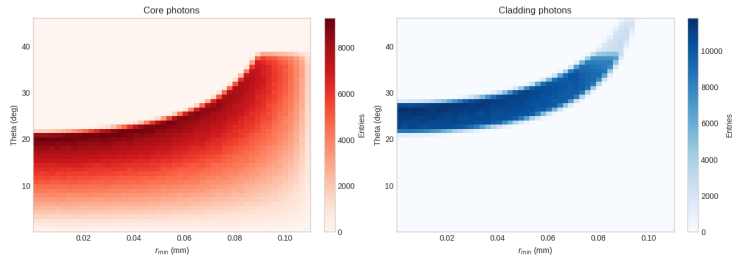


Figure 9: Emission angle θ as a function of the minimum distance from the fibre axis r_{min} for core (left, in red) and cladding (right, in blue) photons.

However, this trapping mechanism is strictly bounded by the condition for total internal reflection. As clearly visible in both distributions, there is a sharp upper cutoff for the angle θ. Photons emitted at longitudinal angles exceeding this r_{min}-dependent threshold strike the fibre's interfaces, and instead of being internally reflected, are refracted out of the fibre and lost. This explains why no photons are detected above this specific boundary: they escape the guiding structure entirely and never reach the photodetector at the fibre's end.

4.4 Intensity Measurement and Attenuation Length

Another objective of this experiment is to evaluate the intensity at different positions and angles, extract the attenuation length of the scintillating fibre, and compare them with the simulation data. To achieve this, intensity measurements were performed at 20 different excitation locations along the fibre (starting from 0 mm to 2000 mm), scanning vertical angles from 0° to 40°. The resulting two-dimensional distribution is presented in Figure 10.

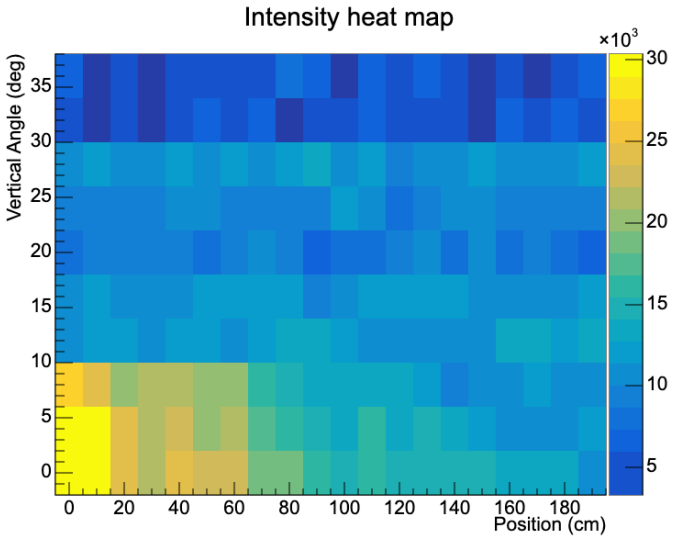


Figure 10: Two-dimensional histogram of the integrated intensity as a function of the excitation position and the vertical angle.

Consistent with the radial symmetry observations (Figure 6), the intensity peaks at vertical angles close to 0° and rapidly decreases at larger angles. Furthermore, the maximum intensity is recorded when the excitation source is closest to the detector ($x \approx 0$ cm) and decays as the distance increases.

To quantitatively evaluate the attenuation length, the integrated intensity of each measurement was extracted as a function of the excitation position for different emission angles (Figure 11) and compared with the simulated data (Figure 12).

As observed in the experimental data, the intensity clearly decays exponentially as a function of position for small angles (between 0° and 8°). In contrast, at larger angles, the intensity profile becomes mostly flat and approaches the background noise level. This occurs because highly skewed photons (large θ) undergo significantly more reflections and travel a longer effective path length, causing them to be heavily attenuated or lost via refraction before reaching the detector. Therefore, to accurately determine the attenuation length of the fibre, we only considered measurements taken at small angles.

In contrast, the simulated profiles behave differently: the curves appear much flatter, and the maximum overall intensity is observed at larger angles, specifically around 20° to 25°. This discrepancy can be attributed to the fundamental difference in the geometric acceptance of the two detection systems. The virtual detector in the simulation has an effectively infinite geometric acceptance, meaning it counts every single photon that successfully exits the fibre. Conversely, the spectrometer used in the laboratory has a strictly limited geometric acceptance due to its narrow entrance aperture. When positioned at 0°,

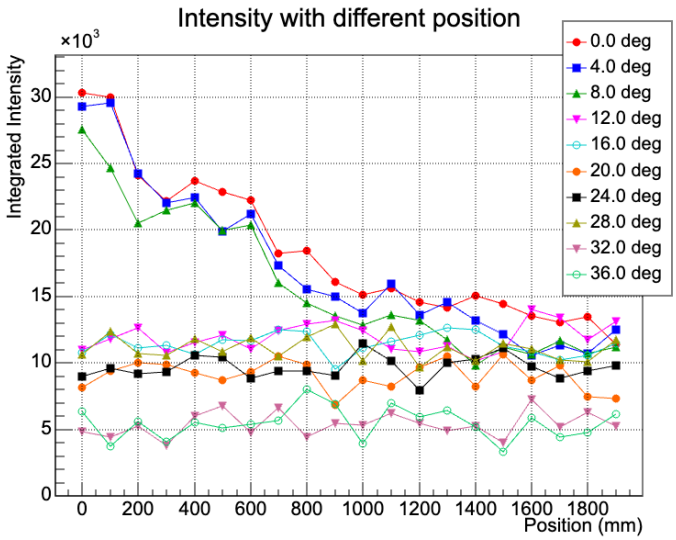


Figure 11: Integrated intensity as a function of the excitation position for different vertical angles.

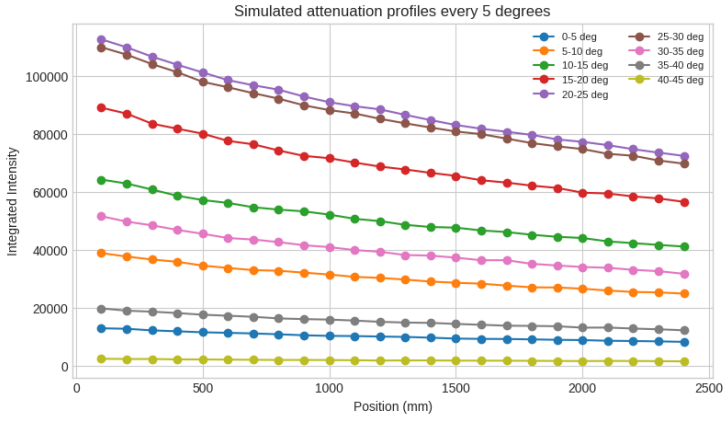


Figure 12: Simulated attenuation profiles for different angular ranges.

it perfectly captures the forward-emitted light. However, at larger angles, the light exiting the fibre becomes more divergent and misses the spectrometer's entrance. Therefore, the instrument fails to detect these off-axis photons, leading to the heavily suppressed and noise-dominated signals observed at higher angles in the experimental data.

To extract the attenuation length, the intensity versus position data for small angles was fitted using a standard exponential decay model:

$$I(x) = I_0 e^{-ax} \tag{6}$$

The fits are shown in Figures 13 and 14.

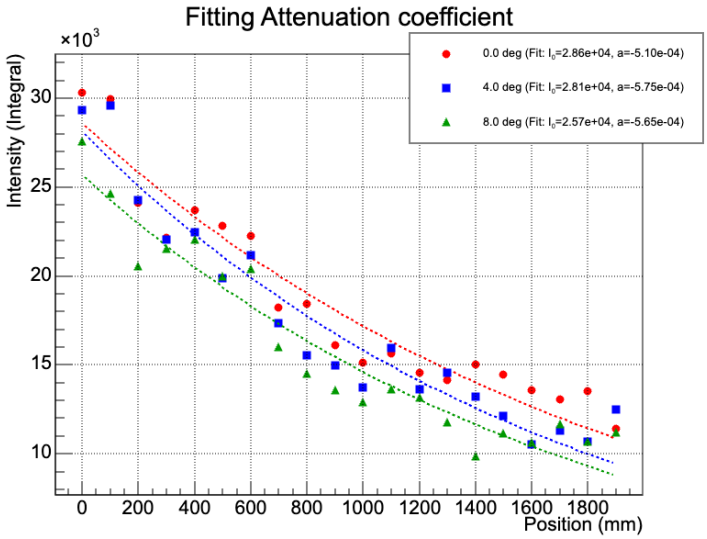


Figure 13: The integrated intensity as a function of position for small emission angles. The dashed lines correspond to the fitting curves.

From these plots, it can be seen that in the actual data the attenuation coefficient slightly increases with the angle due to the additional path length and reflection losses experienced by non-meridional photons. The corresponding attenuation length, Λ, is simply the inverse of the attenuation coefficient:

$$\Lambda = \frac{1}{a} \tag{7}$$

To evaluate the intrinsic attenuation length of the scintillating fibre (minimizing the contribution of reflection losses, which theoretically disappear at 0°) we evaluated the coefficient at the minimal vertical angle. Using the fitted parameter $a = 5.10 \times 10^{-4}$ mm^{-1}, the attenuation length is calculated as:

$$\Lambda = \frac{1}{5.10 \times 10^{-4}} \approx 1961 \text{ mm} = 1.96 \text{ m} \tag{8}$$

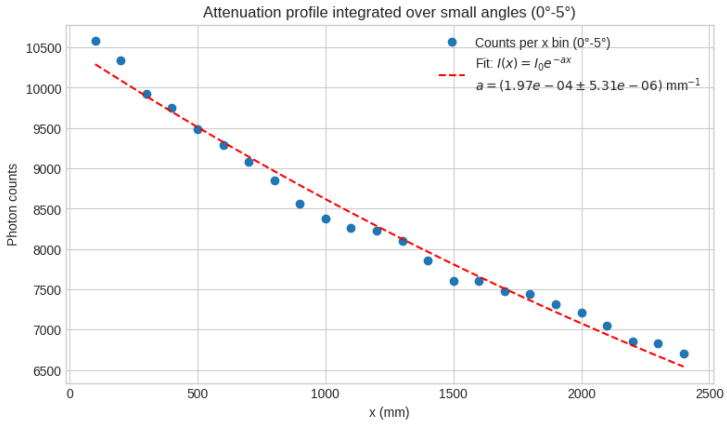


Figure 14: Simulated attenuation profile integrated over small emission angles. The photon counts as a function of the distance x (blue markers) are fitted with an exponential decay model (red dotted line).

If we compare it to the one computed from the simulated data (Figure 14), where we obtained:

$$\Lambda_{sim} = \frac{1}{1.97 \times 10^{-4}} \approx 5076 \text{ mm} = 5.076 \text{ m} \tag{9}$$

This significant discrepancy is expected and can be attributed to the idealized nature of the Monte Carlo simulation. In the Geant4 model, the scintillating fibre is treated as geometrically perfect. In contrast, the actual fibre used in the laboratory is subject to real-world physical imperfections. Furthermore, macroscopic factors such as micro-bendings along the fibre, impurities in the polystyrene core, and potential aging of the plastic material, all contribute to a higher overall absorption coefficient. Consequently, the experimentally measured attenuation length ($\Lambda \approx 1.96$ m) represents a realistic operational value, whereas the simulated result ($\Lambda_{sim} \approx 5.08$ m) represents a theoretical upper limit for an optically perfect fibre.

5 Conclusion

Overall, this study provides a comprehensive analysis of the optical properties and light transport mechanisms of scintillating fibres, focusing on their application within the LHCb experiment after Upgrade I. Understanding the attenuation profile of these fibres is crucial for accurately reconstructing particle trajectories and energies in high-energy physics detectors.

Through experimental measurements, the emission spectrum of the fibre was successfully isolated by applying dark count subtraction, filtering out environmental noise. The analysis of the angular distribution revealed a radial emission

peaking at 0°, though a slight experimental misalignment was identified and accounted for.

By comparing laboratory data with Geant4 Monte Carlo simulations, we validated the theoretical models of light propagation. The simulations highlighted the distinct behaviors of core and cladding photons, specifically demonstrating how non-meridional rays can remain trapped at higher longitudinal angles due to their helical trajectories, bounded by the conditions of total internal reflection.

A critical finding of this study is the marked difference in geometric acceptance between the simulated ideal detector and the real spectrometer, which fundamentally shapes the detected angular intensity profiles. Finally, the attenuation length of the fibre was extracted using the small-angle emission data to minimize reflection losses. The experimental attenuation length was found to be $\Lambda \approx 1.96$ m. When compared to the theoretical simulated value ($\Lambda_{sim} \approx 5.08$ m), the discrepancy highlights the impact of real-world physical imperfections, which are not present in the idealized computational model.

In conclusion, the experiment successfully characterized the scintillating fibre's optical behavior, demonstrating the necessity of combining computational models with experimental validations to accurately assess detector components for the LHCb SciFi tracker.

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